1.14 Specification of lathe

The size of a lathe is specified by the following points

1. The length of the bed.
2. Maximum distance between live and dead centres.
3. The height of centres from the bed.
4. The swing diameter
   a. The swing diameter over bed – It refers to the largest diameter of the work that will be rotated without touching the bed.
   b. The swing diameter over carriage – It refers to the largest diameter of the work that will revolve over the saddle.
5. The bore diameter of the spindle.
6. The width of the bed.
7. The type of the bed.
8. Pitch value of the lead screw.
9. Horse power of the motor.
10. Number and range of spindle speeds.
11. Number of feeds.
12. Spindle nose diameter.
13. Floor space required.
14. The type of the machine

1.15 Work Holding Devices used in a lathe

The work holding devices are used to hold and rotate the workpieces along with the spindle. Different work holding devices are used according to the shape, length, diameter and weight of the workpiece and the location of turning on the work.

They are,

1. Chucks
2. Face plate
3. Driving plate
4. Catch Plate
5. Carriers
6. Mandrels
7. Centres
8. Rests

Chucks

Work pieces of short length, large diameter and irregular shapes, which cannot be mounted between centres, are held quickly and rigidly in chuck. There are different types of chucks namely, three jaw universal chuck, four jaw independent chuck, Magnetic chuck, Collect chuck and Combination chuck.

Three Jaw self-centering chuck

The three jaws fitted in the three slots may be made to slide at the same time by an equal amount by rotating any one of the three pinions by a chuck key. This type of chuck is suitable for holding and rotating regular shaped workpieces like round or hexagonal rods about the axis of the lathe.

Advantages

1. The work is held simply and quickly because of the three jaws are moved to slide at the same time.
2. So, time is saved.
3. It is suitable for apprentice.

Disadvantage

1. Workpieces of irregular shapes cannot be held by this chuck.

Four jaw independent chuck

There are four jaws in this chuck. Each jaw is move independently by rotating a screw with the help of a chuck key. A particular jaw may be moved according to the shape of the work. Hence this type of chuck can hold works of irregular

XII BME_English version CHAPTER 01.indd 16 1/8/2020 1:45:53 PM